Notes: White (Econometrica, 1980)

A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity

Contents

1 Big picture

- **Question.** How can we conduct valid inference in a linear regression when the errors may be heteroskedastic and we do not want to specify a parametric variance model?
- **Main problem.** OLS coefficients remain consistent under exogeneity, but the usual homoskedastic covariance matrix can be inconsistent.
- **Main contribution.** White proposes a heteroskedasticity-consistent covariance matrix estimator using squared OLS residuals and regressors.
- **Modern name.** The estimator is the White robust covariance matrix, or HC0 sandwich estimator.
- **Key insight.** We do not need to estimate every observation-specific error variance; we only need to estimate the average matrix entering the asymptotic variance.
- **Direct test.** The paper derives the auxiliary-regression White test: regress squared residuals on regressors, squares, and cross-products, and use nR^2.
- **Practical takeaway.** Even if heteroskedasticity cannot be fully modeled or eliminated, robust inference remains possible.

2 Model objects and notation

2.1 Linear model and exogeneity

$$Y_i = X_i\beta_0 + \varepsilon_i, \qquad E[X_i'\varepsilon_i] = 0.$$

The observations are independent but not necessarily identically distributed. The model covers fixed regressors, random regressors, and stratified cross sections.

Interpretation. The paper is deliberately broad: heteroskedasticity may depend on covariates, and observations need not be identically distributed.

2.2 OLS and the asymptotic variance object

$$\hat{\beta}_n = (X'X)^{-1}X'Y.$$

Define

$$M_n = n^{-1}\sum_i E[X_i'X_i], \qquad V_n = n^{-1}\sum_i E[\varepsilon_i^2 X_i'X_i].$$

Then the general asymptotic covariance of $\sqrt{n}(\hat{\beta}_n - \beta_0)$ is

$$M_n^{-1}V_nM_n^{-1}.$$

Interpretation. Under homoskedasticity, $V_n = \sigma^2 M_n$. Under heteroskedasticity, this simplification fails.

2.3 White's covariance estimator

Let $e_i = Y_i - X_i\hat{\beta}_n$. White estimates V_n by

$$\hat{V}_n = n^{-1}\sum_i e_i^2 X_i'X_i.$$

The robust covariance estimator is

$$(X'X/n)^{-1}\hat{V}_n(X'X/n)^{-1}.$$

Interpretation. This is the sandwich estimator: bread, meat, bread. The meat is the empirical average of squared residuals times regressor products.

3 Models and theoretical specifications

3.1 Lemma 1: consistency of OLS

$$\hat{\beta}_n \to \beta_0 \quad a.s.$$

Interpretation. Heteroskedasticity does not invalidate point estimates under exogeneity. It invalidates the conventional covariance estimator.

What not to over-read. This result does not imply OLS is efficient or that conventional standard errors are valid.

3.2 Lemma 2: asymptotic normality

$$\sqrt{n}\, M_n V_n^{-1/2} M_n(\hat{\beta}_n - \beta_0) \Rightarrow N(0, I_K).$$

Interpretation. The limiting variance is the robust object $M_n^{-1} V_n M_n^{-1}$, so inference requires estimating V_n.

3.3 Theorem 1: heteroskedasticity-consistent covariance

$$\hat{V}_n - V_n \to 0,$$
$$(X'X/n)^{-1}\hat{V}_n(X'X/n)^{-1} - M_n^{-1} V_n M_n^{-1} \to 0.$$

Interpretation. This is the central theorem. It justifies robust Wald tests, t-type tests, and confidence intervals.

Economic message. The theorem changes heteroskedasticity from a fatal inference problem into a robust-standard-error problem.

What not to over-read. The theorem is asymptotic. It is not a small-sample exact result and does not make OLS efficient.

3.4 Theorem 2: direct covariance-comparison test

Let D_n collect the lower-triangle elements of $\hat{V}_n - \hat{\sigma}_n^2(X'X/n)$. Under the null,

$$nD_n' \hat{B}_n^{-1} D_n \Rightarrow \chi^2_{K(K+1)/2}.$$

Interpretation. The test compares the robust covariance logic with the homoskedastic covariance logic.

What not to over-read. Rejection may reflect heteroskedasticity, but it may also reflect model misspecification or error-regressor dependence.

3.5 Corollary 1: the nR^2 White test

Run the auxiliary regression

$$e_i^2 = a_0 + a_1\psi_{i1} + \cdots + a_{K(K+1)/2}\psi_{i,K(K+1)/2} + u_i,$$

where the ψ variables are the squares and cross-products of the original regressors. Then

$$nR^2 \Rightarrow \chi^2_{df}.$$

Interpretation. This is the familiar applied White test. It operationalizes the covariance-comparison theorem using one auxiliary regression.

What not to over-read. The nominal chi-square reference requires assumptions such as homokurtosis; in finite samples the test can be sensitive to too many auxiliary terms.

3.6 Theorem 3: when heteroskedasticity may not matter

$$\hat{\sigma}_n^2(X'X/n) - V_n \to 0$$

if and only if

$$n^{-1}\sum_i(\sigma_i^2 - \sigma_n^2)(E[X_i'X_i] - M_n) \to 0.$$

Interpretation. Heteroskedasticity affects conventional standard errors only when error variance is systematically related to regressor second moments.

What not to over-read. A non-rejection of the White test is not proof that the errors are literally homoskedastic.

4 Theoretical design and interpretation

4.1 Why no variance model is required

White avoids estimating a variance function. The estimator only needs the average matrix $n^{-1}\sum E[\varepsilon_i^2 X_i'X_i]$.

Interpretation. The paper shifts the task from modeling every variance to estimating the average object needed for inference.

4.2 Why squared residuals are enough

The OLS residuals are close enough to the true errors asymptotically that $n^{-1}\sum e_i^2 X_i'X_i$ estimates the same limit as $n^{-1}\sum \varepsilon_i^2 X_i'X_i$.

Interpretation. Consistent estimation of an average does not require consistent estimation of every component.

4.3 Why the test is also a specification diagnostic

If the mean equation is misspecified, residuals contain specification error and may be related to regressors. The White test can therefore reject because the regression function is wrong.

Interpretation. The test is broad and useful, but broad rejection makes diagnosis cautious.

4.4 Efficiency versus valid inference

Robust standard errors validate inference under heteroskedasticity, but they do not make OLS efficient. If a credible variance model exists, weighted least squares may improve precision.

Interpretation. White's estimator is not anti-modeling; it provides a safe inference baseline and a way to evaluate variance models.

5 Original technical blocks

5.1 Abstract and contribution statement

Read: The abstract states the two core contributions: robust covariance estimation and a direct heteroskedasticity test.

Interpretation. The abstract is where the paper defines its practical target: valid inference without a variance model.

Economic message. The paper is about making regression inference robust to unknown heteroskedasticity.

What not to over-read. The abstract does not say variance modeling is useless; it says inference need not depend on getting that model right.

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A HETEROSKEDASTICITY-CONSISTENT COVARIANCE MATRIX ESTIMATOR AND A DIRECT TEST FOR HETEROSKEDASTICITY

BY HALBERT WHITE[1]

This paper presents a parameter covariance matrix estimator which is consistent even when the disturbances of a linear regression model are heteroskedastic. This estimator does not depend on a formal model of the structure of the heteroskedasticity. By comparing the elements of the new estimator to those of the usual covariance estimator, one obtains a direct test for heteroskedasticity, since in the absence of heteroskedasticity, the two estimators will be approximately equal, but will generally diverge otherwise. The test has an appealing least squares interpretation.

Original abstract block.

5.2 Theorem 1 and the HC covariance estimator

Read: This block defines $\hat{V}_n$ and states the consistency of the robust covariance matrix.

Interpretation. This is the origin of White robust standard errors.

Economic message. It lets researchers use OLS while correcting inference for unknown heteroskedasticity.

What not to over-read. It does not make OLS efficient and it is not a finite-sample theorem.

estimate $n^{-1}\sum_{i=1}^{n} E(\varepsilon_i^2 X_i'X_i)$, an average of expectations. To do this it is not necessary to estimate each expectation separately. Under the conditions given above, a consistent estimator is $n^{-1}\sum_{i=1}^{n} \varepsilon_i^2 X_i'X_1$. Unfortunately, ε_i is not observable; however, ε_i can be estimated by $\hat{\varepsilon}_{in} = Y_i - X_i\hat{\beta}_n$, which leads us to consider the estimator

$$\hat{V}_n = n^{-1}\sum_{i=1}^{n} \hat{\varepsilon}_{in}^2 X_i'X_i.$$

In the fixed regressor case, this amounts to replacing the ith diagonal element of Ω, σ_i^2, with $\hat{\varepsilon}_{in}^2$, the ith squared residual.

With the next condition, the estimator $\hat{V}_n$ becomes the key to solving the problem of obtaining a heteroskedasticity-consistent covariance estimator.

Assumption 4: There exist positive constants δ and Δ such that for all i

$$E(|X_{ij}^2 X_{ik} X_{il}|^{1+\delta}) < \Delta \qquad (j, k, l = 1, \ldots, K).$$

Uniformly bounded fixed or stochastic regressors are sufficient for Assumption 4 to hold.[3]

We now present the first main result.

Theorem 1: (i) $|\hat{V}_n - \bar{V}_n| \xrightarrow{a.s.} 0$ *under Assumptions* 1, 2, 3(a) *and* 4; (ii)

$$|(X'X/n)^{-1}\hat{V}_n(X'X/n)^{-1} - \bar{M}_n^{-1}\bar{V}_n\bar{M}_n^{-1}| \xrightarrow{a.s.} 0$$

under Assumptions 1, 2, 3(a), *and* 4; (iii)

$$n(R\hat{\beta}_n - r)'[R(X'X/n)^{-1}\hat{V}_n(X'X/n)^{-1}R']^{-1}(R\hat{\beta}_n - r) \overset{A}{\sim} \chi_q^2,$$

Original Theorem 1 block.

5.3 Explanation and extensions of Theorem 1

Read: White explains that the estimator supports Wald tests and confidence intervals and can extend to nonlinear and IV models.

Interpretation. The sandwich logic generalizes beyond linear OLS.

Economic message. The practical value is broad: a single covariance principle can be used across many estimators.

What not to over-read. The extensions are not a full modern treatment of clustering, serial correlation, or high-dimensional data.

provides the heteroskedasticity-consistent covariance matrix estimator, $(X'X/n)^{-1}\hat{V}_n(X'X/n)^{-1}$. Using this estimator to test linear hypotheses in the usual way gives correct results asymptotically, as the third part of the theorem demonstrates. This result fills a substantial gap in the econometrics literature, and should be useful in a wide variety of applications. Since the convergence to normality of $\sqrt{n}(\hat{\beta}_n - \beta_0)$ can be shown to be uniform, the heteroskedasticity-consistent covariance estimator can also be shown to be appropriate for use in constructing asymptotic confidence intervals.

In fact, results similar to propositions (i) and (ii) of Theorem 1 were stated over a decade ago by Eicker [**5**], although Eicker considers only fixed and not stochastic regressors. Also, if $r = 0$ and if R is the $1 \times K$ vector with ith element equal to unity and the rest zero, then the χ_1^2 test statistic of (iii) is precisely the square of the asymptotic normal statistic (analogous to the t test) proposed by Eicker [**4**] for the heteroskedastic case in an even earlier classic paper. It is somewhat surprising that these very useful facts have remained unfamiliar to practicing econometricians for

Original explanation block.

5.4 Theorem 2: direct covariance-comparison test

Read: The theorem compares $\hat{V}_n$ with the homoskedastic covariance component.

Interpretation. Large differences indicate the conventional covariance matrix is unreliable under the maintained assumptions.

Economic message. The test diagnoses whether simple homoskedastic standard errors are adequate.

What not to over-read. Rejection can reflect heteroskedasticity or broader specification failure.

$$E(|X_{ij}^4 \Psi_{is} \Psi_{it}|^{1+\delta}) < \Delta \qquad (j = 1, \ldots, K;\ s, t, = 1, \ldots, K(K+1)/2).$$

In vector notation, we can express the lower triangle of the matrix difference $\hat{V}_n - \hat{\sigma}_n^2(X'X/n)$ as the difference vector

$$D_n(\hat{\beta}_n, \hat{\sigma}_n^2) = n^{-1} \sum_{i=1}^{n} \Psi'_{in}[\hat{\varepsilon}_{in}^2 - \hat{\sigma}_n^2].$$

We also use an estimator of $\bar{B}_n$, defined as

$$\hat{B}_n = n^{-1} \sum_{i=1}^{n} [\hat{\varepsilon}_{in}^2 - \hat{\sigma}_n^2]^2 (\Psi_i - \hat{\Psi}_n)'(\Psi_i - \hat{\Psi}_n).$$

With this structure, we can now give a direct test for inconsistency of the usual least squares covariance matrix estimator, $\hat{\sigma}_n^2(X'X/n)^{-1}$.

THEOREM 2: *Given Assumptions* 1, 2(b), 3(b), *and* 5–7, *if ε_i is independent of X_i and $E(\varepsilon_i^2) = \sigma_0^2$ for all i, then*

(1) $$nD_n(\hat{\beta}_n, \hat{\sigma}_n^2)'\hat{B}_n^{-1}D_n(\hat{\beta}_n, \hat{\sigma}_n^2) \overset{A}{\sim} \chi^2$$

Original Theorem 2 block.

5.5 Auxiliary regression for the White test

Read: The auxiliary regression uses squared residuals and all squares/cross-products of regressors.

Interpretation. This converts the matrix test into an OLS regression.

Economic message. The simplicity of this regression explains the test's popularity.

What not to over-read. Too many cross-products can create finite-sample noise and redundant degrees of freedom.

$E(\varepsilon_i^4) = \mu_4$ for all i. Now consider the artificial regression:

$$
\text{(2)} \qquad \hat{\varepsilon}_{in}^2 = \alpha_0 + \alpha_1 \Psi_1 + \alpha_2 \Psi_2 + \ldots + \alpha_{K(K+1)/2} \Psi_{K(K+1)/2}
$$

$$
= \alpha_0 + \sum_{j=1}^{K} \sum_{k=j}^{K} \alpha_s X_{ij} X_{ik} \qquad (i = 1, \ldots, n)
$$

[5] If the investigator is unsure about the correctness of the model's specification, a practical way of proceeding would be to use the statistic (1), and if rejection occurs, apply a specification test of the type proposed by Hausman [**13**] or White [**24**]. These latter tests are sensitive to model misspecification, but not heteroskedasticity. Hence, accepting the null hypothesis of no model misspecification would indicate that rejection of the null hypothesis of Theorem 2 is indeed due to heteroskedasticity, whereas rejection of the null hypothesis of no misspecification indicates that rejection of the null hypothesis of Theorem 2 is due to model misspecification. Since these tests are in general dependent, the formal size of this sequential procedure will be difficult to determine. Note also that both specification tests mentioned above may detect only a lack of independence between errors and regressors, instead of misspecification.

Original auxiliary-regression block.

5.6 Corollary 1: the nR^2 statistic

Read: The corollary gives the textbook nR^2 implementation.

Interpretation. It is the most convenient applied version of the White test.

Economic message. It gives researchers a low-cost diagnostic before trusting conventional covariance estimates.

What not to over-read. The nR^2 reference distribution uses additional assumptions; robust standard errors can still be used without relying on the test.

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where the α's are parameters to be estimated by OLS. Equation (2) directs us to regress the squared estimated residuals on all second order products and cross-products of the original regressors. The next result uses the fact that (1) is asymptotically equivalent to testing the joint hypothesis $\alpha_1 = \alpha_2 = \ldots = \alpha_{K(K+1)/2} = 0$ using the standard R^2 statistic from the regression (2), given the homokurtosis assumption. Formally, we have the following corollary.

Corollary 1: *Given Assumptions* 1, 2(*b*), 3(*b*), 4–6, *if* ε_i *is independent of* X_i, *and* $E(\varepsilon_i^2) = \sigma_0^2$, $E(\varepsilon_i^4) = \mu_4$ *for all i, then*

$$
\text{(3)} \qquad nR^2 \overset{A}{\sim} \chi^2_{K(K+1)/2}
$$

where R^2 *is the (constant-adjusted) squared multiple correlation coefficient from the*

Original Corollary 1 block.

5.7 Theorem 3: low-power and innocuous heteroskedasticity

Read: The theorem characterizes cases in which heteroskedasticity does not make the conventional covariance inconsistent.

Interpretation. Heteroskedasticity matters for OLS inference only when it is aligned with regressor second moments.

Economic message. The White test targets heteroskedasticity that changes inference, not every possible variance difference.

What not to over-read. A small statistic does not prove constant variance.

(2) should be zero (as in the original Hildreth-Houck [**14**] model) or negligible, power (and degrees of freedom) may be gained by omitting some terms in (2). A finite sample investigation of the power of (3) absolutely and relative to existing tests will be undertaken in future work.

Are there cases in which heteroskedasticity does *not* result in inconsistency for $\hat{\sigma}_n^2(X'X/n)^{-1}$? If so, one should expect the power of (1) and (3) to be low in absolute terms for these cases. Such circumstances are easily characterized, although it might sometimes be difficult to detect these cases independently of the statistics given. This characterization is given by the next simple result.

THEOREM 3: *Given Assumptions* 1, 2(*a*) *and* ε_i *independent of* X_i, $|\hat{\sigma}_n^2(X'X/n) - \bar{V}_n| \xrightarrow{\text{a.s.}} 0$ *if and only if*

Original Theorem 3 block.

6 Short summary

- White introduces a covariance estimator that is consistent under heteroskedasticity without modeling the variance function.
- The estimator is $\hat{V}_n = n^{-1} \sum e_i^2 X_i' X_i$ embedded in the sandwich covariance matrix.
- OLS point estimates remain consistent under exogeneity, but conventional standard errors may be wrong.
- Theorem 1 justifies robust Wald tests and confidence intervals.
- Theorem 2 gives a formal covariance-comparison test.
- Corollary 1 gives the practical nR^2 auxiliary-regression White test.
- Theorem 3 explains why some heteroskedasticity may not affect conventional covariance estimation.
- The paper's legacy is the standard use of robust standard errors in empirical work.

7 Expanded conclusion

7.1 Core conclusion

White shows that valid large-sample inference in linear regression does not require a correct parametric model of heteroskedasticity. A residual-based sandwich estimator consistently estimates the covariance matrix needed for OLS inference.

7.2 Econometric intuition

The key idea is that one need not estimate every observation-specific variance. Inference only requires an average matrix involving squared errors and regressor products, and that matrix can be estimated by replacing errors with residuals.

7.3 Practical guidance

Use robust standard errors when heteroskedasticity is plausible. Do not treat robust standard errors as a cure for omitted variables, simultaneity, serial correlation, clustering, or bad functional form. Use cluster or HAC variants when the sampling structure requires them.

7.4 Why the paper became canonical

The paper made robust standard errors a default tool in empirical economics and introduced the broader covariance-estimator comparison logic underlying many modern specification tests and sandwich estimators.

7.5 Limits and modern perspective

The original paper is an asymptotic independent-observation theory. Modern work extends the same sandwich idea to leverage corrections, clustered dependence, serial correlation, GMM, nonlinear estimators, and bootstrap inference.

7.6 One-sentence takeaway

White's core lesson is that unknown heteroskedasticity need not invalidate regression inference if the researcher estimates the covariance of the moment conditions directly.